

Encrypted Traffic Classification Based on Two-Phase Feature Fusion

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Abstract—The existing encrypted traffic classification schemes have not taken full advantage of the principles of network traffic encryption technology, resulting in poor classification accuracy and interpretability. In fact, the transmission process of encrypted flows can be divided into a handshake phase and a secure data transmission phase, each with distinct characteristics and inter-packet correlations. Therefore, an effective approach for encrypted traffic classification should employ different processing methods for each phase and then combine the results. In this paper, we propose the Two-Phase Feature Fusion (2P2F) approach for classifying encrypted network flows. In 2P2F, during the handshake phase, a convolutional neural network (CNN) processes the encryption protocol extension field to extract application-related features. For the secure data transmission phase, a transformer model captures content-related features from the TCP segment length sequence. Finally, an attention mechanism is used to fuse the refined features from both phases, allowing the classification model to focus more on important features and achieve accurate classification of encrypted network flows. The experimental results demonstrate the superiority of the proposed scheme.

Index Terms—Encrypted traffic classification, two-phase features, feature fusion, deep learning.

I. INTRODUCTION

WITH the rapid development of the Internet technology and its applications, network traffic is constantly increasing, which puts a strain on network management. Effective network traffic classification is essential for enhancing network management capabilities, improving service quality,

and ensuring application security [1], [2], [3]. However, as the demand for data privacy and security keeps growing, an increasing proportion of network traffic is encrypted, making classification more challenging [4], [5], [6].

Traditional network traffic classification approaches, such as port-based, deep packet inspection (DPI) based, and machine learning-based, are ineffective in classifying encrypted network traffic. Port-based approaches rely on known ports assigned by the Internet Assigned Numbers Authority (IANA), rendering them incapable of handling dynamic or disguised ports [7]. DPI-based approaches analyze the payload data of the entire data packet to match predefined regular expressions, but they cannot inspect encrypted packet payloads [8]. Machine learning-based approaches heavily rely on feature engineering and professional knowledge, which can be a challenging and laborious task [9], [10], [11], [12], [13].

Recently, deep learning (DL) has gained significant attention as a potential solution for encrypted network traffic classification due to its capability to effectively extract complex features and capture intricate relationships automatically [14]. However, many existing deep learning-based approaches tend to focus solely on either classifying the application types [15], [16], [17], [18], [19] or content types [20], [21], [22] of encrypted traffic. Furthermore, these approaches often exhibit limitations, such as either solely considering the characteristics of individual packets rather than the packet sequences within network flows, or failing to address the underlying principles of network encryption protocols.

The most widely used encryption protocols are Secure Sockets Layer (SSL) and its successor Transport Layer Security (TLS) protocol [23]. According to the workflow of SSL/TLS, an encrypted network flow can be naturally divided into two phases, namely, the TLS handshake phase and the secure data transmission phase [24], [25]. These two phases expose different observable information and exhibit distinct inter-packet correlations. In the handshake phase, packets follow a well-defined protocol format and contain unencrypted handshake metadata that reflects negotiation details between the client and server. In the secure data transmission phase, the application payload is encrypted and remains opaque. Nevertheless, traffic-analysis side-channel patterns such as the sequence and distribution of TCP segment lengths can still be observed and may correlate with coarse content types. Importantly, we do not decrypt or inspect application payload contents at any stage. Therefore, to improve classification

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accuracy and interpretability, a sensible strategy is to extract complementary features from these two phases using phase-appropriate models, and then fuse them to obtain the final classification result.

However, many existing approaches treat an encrypted flow as a single homogeneous entity and apply one generic model to the entire flow, or they focus on coarse intra-packet divisions (e.g., header/body) rather than the protocol-phase workflow. As a result, they may fail to fully exploit the heterogeneous yet complementary information available in the handshake metadata and in the transmission-phase traffic patterns.

In this paper, we present a novel encrypted network traffic classification scheme called Two-Phase Feature Fusion (2P2F). This scheme is motivated by the analysis that different phases of the encrypted network flows play distinct roles in traffic classification. In 2P2F, during the handshake phase, a convolutional neural network (CNN) processes the encryption protocol extension field to extract application-related features. Meanwhile, in the secure data transmission phase, a transformer model captures side-channel statistical features from the TCP segment length sequence. Finally, the attention mechanism is employed to fuse refined features from both phases, allowing the model to focus more on important features, effectively classify encrypted network data flows.

The contributions of this paper can be summarized as follows:

- We propose an encrypted traffic classification framework based on two-phase feature fusion. In this framework, features are extracted from both the handshake phase and the secure data transmission phase, which are subsequently fused to achieve accurate classification. This scheme leverages the unique characteristics of both phases, enabling precise classification results while utilizing only a limited number of features.
- We design two feature extraction schemes tailored to the characteristics of each phase. The CNN-based scheme is designed to process the encryption protocol extension field of handshake phase to extract application-related features, while a transformer model is employed to capture content-related features from the TCP segment length sequence in the secure data transmission phase.
- We design an attention mechanism based feature fusion algorithm for the semantic features from the handshake phase with the statistical features from the secure data transmission phase of encrypted network flows. This approach effectively classifies encrypted network data by focusing on the most important features.
- We implement an encrypted traffic classification testbed and construct a real-world network traffic dataset. Through extensive experiments, we compare the performance of our proposed scheme against state-of-the-art methods. The experimental results demonstrate the superiority and effectiveness of the proposed scheme.

The remainder of this paper is organized as follows. Section II describes some of the related work of network traffic classification. Section III introduces the concept of encrypted traffic classification problem and motivation. Then, the two-phase feature fusion-based encrypted traffic

classification scheme is proposed in Section IV. The experiment results and analysis are given in Section V, and the last Section VI gives the conclusion of this paper.

II. RELATED WORK

In this section, we provide an overview of prior research on traffic classification, categorized into traditional, machine learning-based, and deep learning-based approaches.

A. Traditional Traffic Classification

Port-based schemes [7] encounter challenges in scenarios involving dynamic ports, while DPI-based approaches [26] prove ineffective in classifying encrypted traffic. Protocol analysis-based methods extract the distinctive behavioral characteristics of specific applications by analyzing their communication processes. Korczyński and Duda [15] proposed first-order homogeneous Markov model fingerprints to characterize the dynamics of SSL/TLS sessions. Shen et al. [16], [17] extended it to the second-order Markov model by clustering the certificate length and the first packet length to improve the classification performances. However, due to the caching mechanism, not all encrypted flow contains a certificate.

Pan et al. [18] combined Message Type Sequence (MTS) and the lengths of the first several packets in a network flow to construct the Markov model and hidden Markov model (HMM) and then used a weighted ensemble classifier to improve the results. Despite achieving commendable classification performance, this approach has limitations due to the potential lack of clear distinctions in packet length when different applications transmit identical business content.

In summary, the failure of these traditional methods to reliably handle encrypted payloads necessitates the shift towards more advanced, learning-based approaches.

B. Traditional Machine Learning Based Traffic Classification

Anderson et al. [9] employed logistic regression with a combination of flow metadata, packet length distributions, time distributions, byte distributions, and unencrypted TLS header information to identify malware-encrypted traffic. Sun et al. [10] applied TLS pattern matching to filter out all TLS traffic, followed by Naïve Bayes classification, achieving a classification precision of 93%–96% for HTTPS and TOR traffic.

Sivanathan et al. [11] effectively identified encrypted traffic, particularly IoT devices, by leveraging carefully selected features through the Random Forest algorithm. Cao et al. [12] proposed a real-time accurate SVM training model. They used PCA to extract features from the data to reduce the dimension of the original features and a particle swarm optimization algorithm to search the optimal working parameters of the kernel function automatically. This method can improve traffic classification accuracy through a small number of training samples compared with the traditional SVM. Lashkari et al. [13] used time-related features such as the duration of the flow, flow bytes per second, forward and backward inter-arrival time, etc., to characterize the network traffic using k -nearest neighbor (k -NN) and C4.5 decision tree algorithms.

However, the performance of machine learning methods depends on the rich experience, professional knowledge, and features that are artificially selected.

C. Deep Learning Based Traffic Classification

Deep learning has emerged as a promising technique for encrypted traffic classification. Wang [27] applied a simple SAE automatic encoder to traffic classification. Li et al. [28] introduced the recurrent neural network to network traffic classification and designed the byte segment neural network where the segments of payloads are put into the attention encoder to get the features. Then the softmax classifier is used for classification.

References [20] and [29] designed a novel method that combined the convolutional network and the recurrent network to improve the accuracy of the classification results. The convolutional network extracts the packet features for a single packet. The recurrent network is trained to pick out the flow features based on the inputs of the packet features of any three consecutive packets in a flow. Liu et al. [19] considered the length of packets in the encrypted traffic and proposed an end-to-end model taking a multi-layer bi-GRU encoder to learn the representation of the flow sequence, and reconstructing the original sequence with a multi-layer bi-GRU decoder.

Recently, more advanced deep learning models, such as graph neural networks (GNNs) and transformers, have been widely adopted for encrypted traffic analysis. For instance, researchers have utilized GNNs to capture complex flow interaction patterns for encrypted malware detection [30], [31]. Some works have even begun to differentiate the physical components of a packet, such as the header and payload, for more fine-grained classification [32], [33]. Concurrently, Transformer-based models have demonstrated their potential by applying self-attention mechanisms to traffic sequences [34], [35].

However, existing deep learning-based approaches often neglect the intricate working mechanism and distinctive characteristics of encrypted network protocols, or apply a single, generic model to the entire flow, failing to optimally extract the heterogeneous features from different phases.

III. PRELIMINARIES

In this section, we first investigate the mechanisms of encrypted protocols, then define the problem of encrypted traffic classification, and finally present our research motivation.

A. Encryption Protocol Analysis

As shown in Fig. 1, the interaction process of a typical encryption protocol, such as SSL/TLS, consists of two distinct phases: the handshake phase and the secure data transmission phase. In the handshake phase, we use only observable handshake metadata. In the secure data transmission phase, payload contents are encrypted and are not inspected; we use only side-channel statistical feature, represented by the TCP segment length sequence. This phase-wise design avoids any payload decryption while leveraging complementary cues from protocol metadata and transmission behaviors.

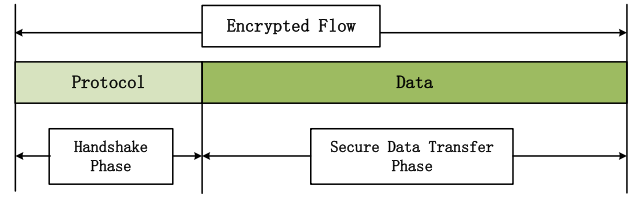


Fig. 1. The composition of encrypted flow.

In the handshake phase, certain TLS handshake metadata (e.g., message fields and extension information) is observable and follows a well-defined protocol format. For a specific network application, these data possess distinct meanings and represent the distinct status and characteristics of a flow. As a result, they serve as valuable features for application-type classification.

On the other hand, the secure data transmission phase involves the transmission of content segmented into a sequence of encrypted packets [36]. Although the packets in this phase are encrypted, different content types (e.g., video streaming, text, or digital files) exhibit discernible behavioral characteristics, including packet size distribution, packet arrival interval distribution, and other statistical features. These features can be effectively utilized for content-type classification of network flow.

B. Problem Definition

Encrypted traffic classification is generally performed to identify the application and/or content type of packet sequence within a given network flow. Specifically, the application type refers to application-related information such as Baidu, a website search engine and Bilibili, an online video platform, or other network applications, while the content type refers to the nature of the transmitted data, such as video, image, text, and so on. To be precise, we define three distinct classification tasks which will be evaluated: “Application-Type Classification” (identifying only the application, e.g., “Baidu”), “Content-Type Classification” (identifying only the content, e.g., “video”), and “Application-Content Classification” (the joint and granular task of identifying the specific pair, e.g., “Baidu-video”) the classification is conducted at three levels: packet level, stream level, and session level, depending on the specific classification requirements or techniques employed for network traffic classification.

The aim of this paper is to propose a methodology for jointly classifying the application and content types of encrypted network flows at the session level, specifically those utilizing the SSL/TLS protocol, as they traverse network nodes including switches or gateways. In this context, a network flow is defined as a series of packets that share the same five-tuple (Source IP, Source port, Destination IP, Destination port, Transport layer protocol) information.

C. Motivation

As discussed in Section III-A, a typical encrypted network flow consists of two distinct phases: the handshake phase and

the secure data transmission phase, each exhibiting unique packet sequence characteristics. Unfortunately, previous studies often treat the encrypted network data flow as a whole entity, overlooking the distinctions between these two phases. This oversight has resulted in poor classification accuracy.

To address the aforementioned issue, we believe that a reasonable solution should *utilize distinct processing methods for different phases of encrypted network flows and then fuse the results to achieve accurate classification*. Specifically, for the handshake phase, since each field in this phase is unencrypted and holds a specific meaning, we propose to employ a CNN model to analyze the handshake phase and extract application-type-relevant features.

The secure data transmission phase relies on encryption technology, rendering the classification of content through the inspection of payload contents infeasible. Nevertheless, statistical features derived from the packet sequence within this phase can be leveraged for content-type classification. To this end, we propose the adoption of a transformer model to analyze the statistical features of the packet sequence, including packet size distribution, with the aim of extracting content-type-relevant features.

Moreover, as web browsing typically generates multiple network flows corresponding to various web components with different types (e.g., video, image, text, etc.), the accuracy of application-content classification can be enhanced by combining the application-type-relevant features and content-type-relevant features extracted from these two distinct phases.

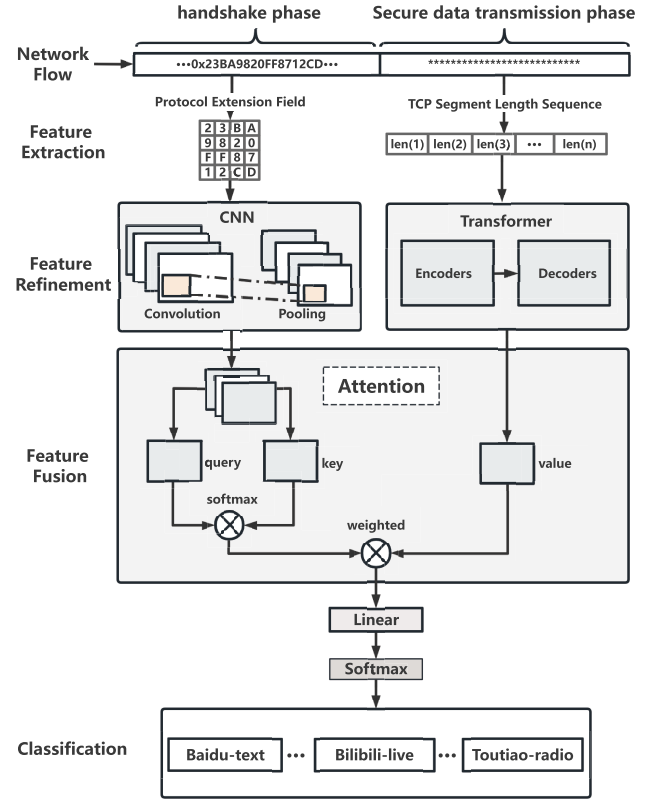


Fig. 2. Overview of the proposed two-phase feature fusion traffic classification.

IV. TWO-PHASE FEATURE FUSION BASED ENCRYPTED TRAFFIC CLASSIFICATION

This section introduces the overview and detail of the proposed encrypted traffic classification scheme.

A. Overview

The architecture of two-phase feature fusion-based encrypted traffic classification is shown in Fig. 2, which consists of the modules of traffic preprocessing, feature extraction, feature refinement, and feature fusion.

In the traffic preprocessing module, the first step involves partitioning the raw traffic into session flows based on their five tuples. Following this, in the feature extraction module, features are extracted and normalized to mitigate the influence of variations in magnitudes. To refine the quality of the extracted features, a CNN model is employed to process the features from the handshake phase, while a transformer model handles the features from the secure data transmission phase. Finally, the refined features are fused using an attention mechanism [37], facilitating the classification of application-content types and yielding the final classification results.

Furthermore, it is worth noting that the CNN model responsible for processing features obtained from the handshake phase and the transformer model responsible for processing features extracted from the secure data transmission phase are pre-trained offline in advance.

B. Feature Extraction

We conduct experiments to evaluate various features extracted from both phases and select two highly effective features, namely the protocol extension information in the handshake phase and the TCP segment length sequence in the secure data transmission phase. Features susceptible to network conditions, such as packet inter-arrival time, are deliberately excluded. Additionally, using IP addresses directly for classification is not a good choice due to the existence of content delivery network (CDN) or other load balancing mechanisms.

1) *Protocol Extension Field*: The extension fields in the SSL/TLS protocol are used to transmit additional information and functionality during the handshake process. For different applications, there are significant differences in the specific negotiation content and the encryption suite between the client and server which are recorded in the protocol extension fields of the SSL/TLS layer. Thus, we extract the raw extension field data from the ‘Client Hello’ and ‘Server Hello’ messages during the handshake phase and convert it to hexadecimal format, which facilitates its transformation into grayscale images. The server name field in the protocol extension field can be forged, so we replace its content with 0.

2) *TCP Segment Length Sequence*: During the secure data transmission phase, content is encrypted to ensure confidentiality, rendering the analysis of the encrypted content ineffective in obtaining useful information about the network flow.

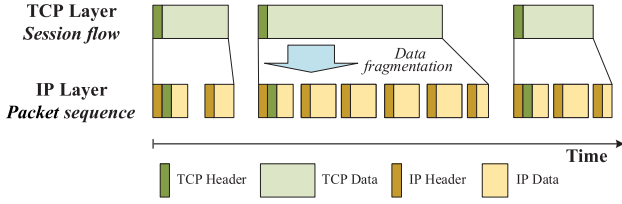


Fig. 3. The mapping between TCP session flow and IP packet sequence.

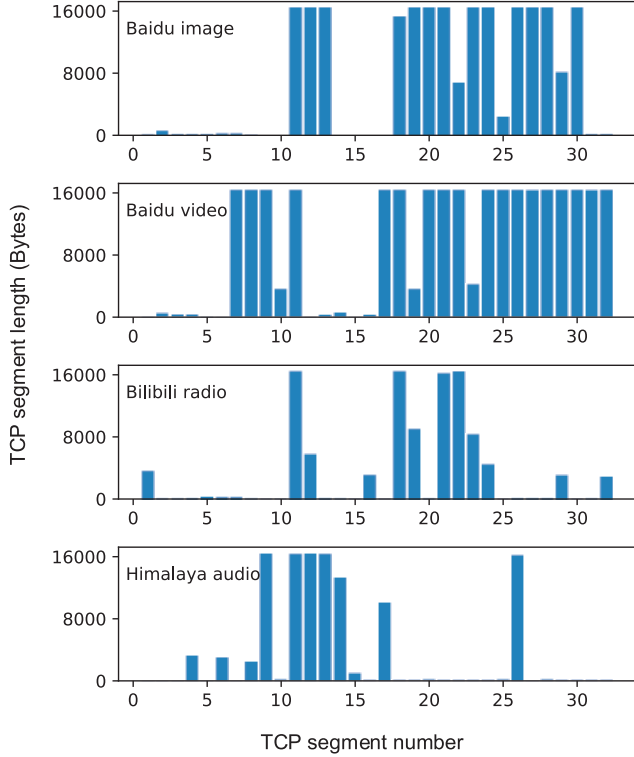


Fig. 4. Distributions of the TCP segment lengths for different types of content.

However, even in the presence of encryption, it is still possible to differentiate between various types of transmitted content, such as text or video, based on the statistical distribution of the packet sequence within the flow. This statistical distribution can serve as a distinctive feature for classifying encrypted network flows.

As illustrated in Fig. 3, the data segments of the TCP session flow are fragmented into a sequence of IP packets, with most packets having uniform sizes. This fragmentation leads to the partial loss of application-related information within the IP packet sequence. To address this, we adopt a different approach compared to previous research that primarily focused on the statistical distribution of packets in the IP layer. Instead, we use Scapy¹ to parse the handshake layer and record layer protocol types of all TLS packets, which are unencrypted, and divide the handshake phase and secure data transmission phase of the encrypted flow based on them. For packets that have been split due to their length exceeding the IP layer limit, we merge them and extract the original TCP segment length

¹<https://scapy.net>

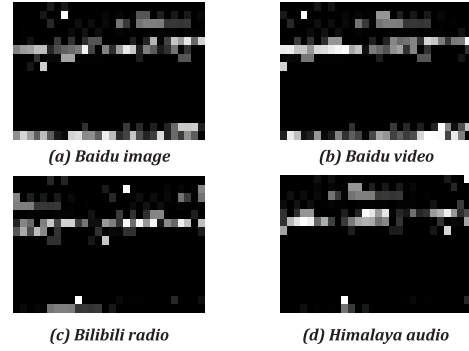


Fig. 5. Grayscale images converted from the features of the extension field for different types of content.

TABLE I
PARAMETERS OF CNN MODEL USED FOR FEATURE REFINEMENT
OF HANDSHAKE PHASE

Parameter	Value
Input image size	$16 \times 28 \times 1$
Convolution layer 1	16maps, 5×5 , Stride:1
Pooling layer 1	max pooling, 2×2 , Stride:1
Convolution layer 2	32maps, 5×5 , Stride:1
Pooling layer 2	max pooling, 2×2 , Stride:1
Activation Function	ReLU
Optimizer	Adam

sequence. Finally, We select the length information of the first 32 TCP data segments in the secure data transmission phase as the feature for classifying encrypted traffic.

Fig. 4 showcases an example histogram illustrating the distribution of the first 32 TCP segment lengths during the secure data transmission phase for four different types of content. It is evident that the transmission of different contents exhibits distinct distributions, further highlighting the potential of utilizing this feature for content classification.

C. Feature Refinement

1) *Feature Refinement for Handshake Phase:* According to the standard of encryption protocols, the data in each fixed position in the protocol extension field denotes a specific meaning, so the data in the protocol extension field has a certain regularity, and this regularity can be analyzed by using the CNN model. Crucially, this 2D image conversion is applied only to the unencrypted handshake field, as it preserves the spatial structure of the protocol data and creates an information-dense (non-sparse) representation for the CNN to analyze.

In order to preserve the information of the protocol extension field in its entirety and also to facilitate the input of this information into the CNN model, we convert the extension field into grayscale image format. The length of extended fields varies depending on the data type, but most will not exceed 448 bytes, we transform the protocol extension field into a 16×28 grayscale image, as shown in Fig. 5. The structure and parameters of the CNN model are shown in Table I.

2) *Feature Refinement for Secure Data Transmission Phase:* The specific content of the data transmission phase is encrypted, but we can still extract sequence features to predict

TABLE II

PARAMETERS OF TRANSFORMER MODEL FOR FEATURE REFINEMENT OF SECURE DATA TRANSMISSION PHASE

Parameter	Value
Input sequence length	32
Embedding dimension	16
Number of encoder layers	4
Number of decoder layers	4
Number of attention heads	8
Feedforward network dimension	128
Dropout rate	0.1
Activation function	ReLU
Optimizer	Adam

user behavior, which are more suitable to be analyzed by using an transformer model. As a special type of RNN, Transformer model not only has the ability for neurons to directly affect themselves at the next moment but also solves the limitation of ordinary RNNs by adding gating mechanisms within the network, to train TCP segment length sequences. The structure and parameters of the model are shown in Table II.

D. Feature Fusion

For the classification of encrypted network flow, the features of different phases of encrypted flow play a certain role in different classification tasks. Feature fusion is to fuse the features extracted and refined from different phases together to achieve more accurate classification results.

In feature fusion, we employ an attention mechanism in our encrypted traffic classification scheme. The attention mechanism allows our approach to selectively focus on the most informative features from each phase of the encrypted flow during the fusion process. It assigns different weights or importance to the features based on their relevance to the classification task at hand. The attention mechanism helps us overcome the unbalanced role of features from different phases by dynamically adjusting the contribution of each feature during the fusion process. It allows us to effectively combine the relevant information from different phases and enhance the overall classification performance. Suppose the output of the CNN model is F_c , and the output of the transformer model is F_t , we use them separately to generate the query, key, and value:

$$Q = W_Q \cdot F_c, \quad (1)$$

$$K = W_K \cdot F_c, \quad (2)$$

$$V = W_V \cdot F_t, \quad (3)$$

where W_Q , W_K and W_V is the learnable parameter matrix. Then we use them to calculate attention scores:

$$F_{\text{fused}} = \text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (4)$$

where d_k is the dimension of the key.

Finally, the fused features are input into the classifier to obtain the traffic label:

$$y = \arg \max_k (\text{softmax}(W_{\text{cls}} \cdot F_{\text{fused}} + b)), \quad (5)$$

where W_{cls} and b is the learnable parameter, k is the index of classes.

V. EVALUATION

This section provides an overview of the dataset and experimental setup utilized in the evaluation, followed by a comprehensive analysis of the experimental results.

A. Experimental Setting

1) *Datasets*: The first dataset in our experiments, 2P2F Website Dataset, was captured by ourselves on the China Environment for Network Innovations (CENI-HeFei) platform [38]. it consists of five commonly used applications, including Bilibili, TouTiao, Himalaya, Netease cloud, and Baidu, and these traffic data are categorized into five types: image, audio, radio, live video, and video. The figures in the table represent the number of session flows, with a total size of 175.8Gb.

In addition, we conducted experiments on the publicly available dataset CIC IoT 2022 [39] and mobile app dataset [40] to validate the generality of our approach. We selected 10 use cases with larger sample sizes from these two datasets for experimentation. The ratio of training and testing data for all datasets is 8:2, and the specific composition is shown in the table III below.

2) *Preprocessing*: We divided the Pcap file that contains all captured packets into multiple Pcap files based on the transport layer protocol type (TCP or UDP) using SplitCap.² We then selected the largest SSL/TLS session flow as our classification object, after dividing the Pcap file generated by accessing a specific website into multiple session flows. This approach reduces the impact of background traffic and ensures the flow is related to the content type.

3) *Experiment Setup*: All these experiments use the Intel 6142 server equipped with 16 core CPU as the hardware platform, a NVIDIA Tesla P4 GPU as the training and testing of the model, with 512G memory and 256G SSD. The software is based on the depth model framework Pytorch 2.0, with cuda version 11.8 and operating system Ubuntu 16.04.

B. Comparison Algorithms

We compare our proposed scheme with three state-of-the-art solutions. These baselines were selected to be representative of the primary categories of classification methods: classic machine learning (CCIT), deep sequence models (FS-Net), and modern semi-supervised fingerprinting (FlowPrint).

- CCIT [11] is a classical method that utilizes a carefully selected set of features to accurately identify encrypted traffic originating from IoT devices. This method employs the random forest algorithm to achieve effective classification.
- FS-Net [19] considered the length of the first n packets in the encrypted data flow and proposed an end-to-end model taking a multi-layer bi-GRU encoder to learn the representation of the flow sequence and reconstruct the original sequence.
- FlowPrint [41] is a semi-supervised approach for fingerprinting mobile apps from encrypted network traffic. It can automatically find temporal correlations among

²<https://www.netresec.com/?page=SplitCap>

TABLE III
THE COMPOSITION OF THREE DATASETS IN TERMS OF THE NUMBER OF SESSION FLOWS

2P2F Website		CIC IOT 2022 [39]		Mobile App [40]	
Application	Count	Device	Count	Application	Count
Baidu video	2030	Amazon Alexa Echo Dot 2	11079	360 cloud	24406
Baidu image	2036	Amazon Alexa Echo Spot	11079	WiFi Master key	6692
Bilibili live video	1094	Amazon Alexa Echo Studio	11187	Douyin	4966
Bilibili radio	2592	Philips Hue Bridge	1094	Lianjia	3615
Toutiao image	11977	Ring Base Station AC1236	791	MangoTV	3272
Toutiao live video	4221	Smart Board	936	Pinduoduo	1733
Himalaya audio	1094	Arlo Base Station	4221	Qutoutiao	2585
Himalaya radio	2592	Netatmo Camera	525	TaoBao	2519
Netease cloud audio	11977	Sonos One Speaker	2592	TanTan	2928
Netease cloud video	4221	Google Nest Mini	1098	ZiRu	2800

TABLE IV
PERFORMANCE OF 2P2F AND COMPARISON
ALGORITHMS ON THREE DATASETS

Algorithm	2P2F Website			CICIoT 2022			Mobile App		
	PR	RC	F1	PR	RC	F1	PR	RC	F1
CCIT	0.87	0.87	0.87	0.99	0.99	0.99	0.81	0.80	0.80
FS-NET	0.84	0.84	0.84	0.93	0.93	0.93	0.93	0.93	0.93
FLOWPRINT	0.79	0.79	0.79	0.45	0.43	0.43	0.81	0.80	0.81
DE-GNN	0.89	0.89	0.89	0.75	0.73	0.73	0.91	0.91	0.91
2P2F-fuse	0.96	0.96	0.96	0.99	0.99	0.99	0.97	0.97	0.97
2P2F-handshake	0.85	0.83	0.82	0.98	0.98	0.98	0.94	0.94	0.94
2P2F-transmission	0.90	0.90	0.90	0.97	0.97	0.96	0.93	0.93	0.93
2P2F-assoc5	0.95	0.91	0.93	0.99	0.98	0.98	0.96	0.95	0.95

destination-related features of network traffic and use these correlations to generate app fingerprints.

- DE-GNN [32] is a state-of-the-art method for fine-grained encrypted traffic classification. This model uses a dual embedding layer to encode the packet header and payload separately, then constructs the data flow into a Traffic Interaction Graph (TIG), and utilizes Graph Neural Networks (GNN) to extract flow-level features.

C. Classification Results With Compared Algorithms

In this evaluation, the *2P2F-fuse* is the complete version of the proposed method, which combines features from two phases to achieve a joint classification of application and content types. The *2P2F-assoc5* signifies the utilization of features from the handshake phase to determine the application type, and the utilization of features from the secure data transmission phase to determine the content type, followed by the simple concatenation of the two results. The *2P2F-handshake* and *2P2F-transmission* respectively represent the direct recognition of application and content types using features solely from the handshake phase and secure data transmission phase.

As shown in Table IV, *2P2F-fuse* shows excellent performances on encrypted traffic classification. It is observed that the proposed 2P2F method has significant improvements.

The results in Table IV demonstrate the superiority of our approach. We first analyze our own model variants as an ablation study to justify our design, addressing the

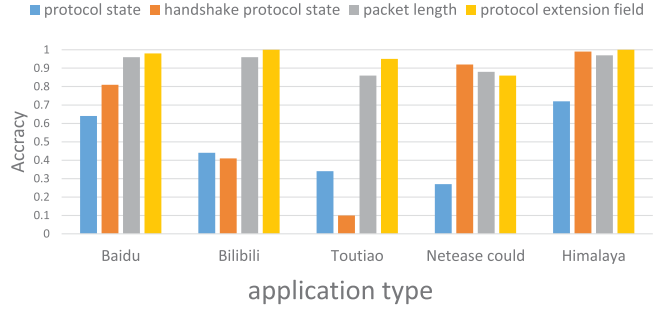


Fig. 6. The accuracy of application type classification among features.

need to isolate each phase's contribution. On the 2P2F Website dataset, *2P2F-handshake* (handshake-only) and *2P2F-transmission* (transmission-only) achieve F1-scores of 0.82 and 0.90, respectively. Our full 2P2F-fuse model significantly outperforms both, with an F1-score of 0.96. This explicitly validates our core thesis that both phases contribute unique, essential information, and our fusion mechanism is necessary to achieve the best results.

Compared to the baselines, this optimized 2P2F-fuse model also shows clear advantages. The baseline methods falter because they fail to properly distinguish and process the two distinct phases of encrypted traffic. CCIT's manual features lack generalization. FS-Net's "one-size-fits-all" model ignores the differing semantics of handshake and transmission packets. FlowPrint's narrow fingerprinting fails in diverse scenarios, and DE-GNN's physical-based header/payload division is less effective than 2P2F's protocol-based (handshake/transmission) split. The failure of these methods further validates that processing each phase with specialized models and fusing the results is crucial for accurate classification.

D. Effectiveness of Our Selected Feature

To evaluate the effectiveness of our selected features, we compared the protocol extension field, TCP segment length sequence, and other features to test their effectiveness in identifying application types and content types, respectively.

It can be observed from Fig. 6 that considering only the handshake protocol state is better than using the full protocol

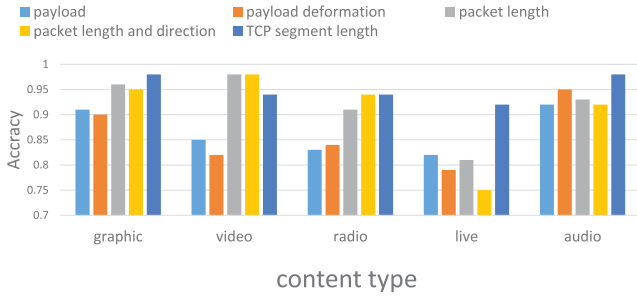


Fig. 7. The accuracy of content type classification among features.

TABLE V

THE PERFORMANCE OF APPLICATION TYPES BASED ON TWO-PHASE FEATURE FUSION

Types	2P2F-fuse-app			2P2F-handshake-app			2P2F-transmission-app		
	PR	RC	F1	PR	RC	F1	PR	RC	F1
Baidu	0.99	0.98	0.99	0.85	0.98	0.91	0.84	0.98	0.90
Bilibili	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.99	0.99
Toutiao	0.99	1.00	0.99	1.00	0.95	0.97	0.99	0.86	0.92
Netease cloud	0.99	0.99	0.99	0.98	0.86	0.92	0.98	0.95	0.96
Himalaya	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.99	1.00
Average	0.99	0.99	0.99	0.97	0.96	0.96	0.96	0.96	0.95

state. Therefore, it is more reasonable to consider different types of features for different phases of the flow. Although the application type classification can be achieved to some extent by using the handshake protocol state, the results show that the classification accuracy of Bilibili and Toutiao is significantly lower compared to the experiment results based on packet features. This also indicates the inability to achieve the best classification result by using features from only one phase of the flow.

As shown in Fig. 7, although we can observe that the classification of content types based on encrypted payload is effective to some extent through comparison experiments, the classification results do not change much if we change all payload data to 255. Therefore, we believe that the real information processed by the CNN should be the contour information constructed by different packet payload lengths in the grayscale image. This contour information also verifies that the length should be considered in encrypted traffic analysis rather than the specific content after encryption.

E. Effectiveness of Feature Fusion

1) *Effectiveness of Feature Fusion for Single Task*: We now study the classification by using different sets of features in order to understand the impact of each feature in classification performance. We study three scenarios: using the fused features from both phases, using the feature only from the handshake phase, and using the feature only from the secure data transmission phase.

The application-type classification results are shown in Table V. It can be observed that the extended field information in the handshake phase is more suitable for application classification than the length information of the TCP segment in the

TABLE VI
THE PERFORMANCE OF CONTENT TYPES BASED ON TWO-PHASE FEATURE FUSION

Types	2P2F-fuse-con			2P2F-handshake-con			2P2F-transmission-con		
	PR	RC	F1	PR	RC	F1	PR	RC	F1
video	0.99	0.99	0.99	0.82	0.89	0.85	0.98	0.94	0.96
radio	0.97	0.94	0.95	0.83	0.66	0.74	0.95	0.94	0.94
image	0.99	0.99	0.99	1.00	0.97	0.98	0.92	0.98	0.95
audio	0.99	1.00	0.99	0.86	0.72	0.78	0.98	0.98	0.98
live	0.93	0.97	0.95	0.74	0.97	0.84	0.93	0.92	0.92
Average	0.97	0.98	0.97	0.85	0.84	0.84	0.95	0.95	0.95

TABLE VII

THE PERFORMANCE OF APPLICATION-CONTENT TYPE CLASSIFICATION BASED ON TWO-PHASE FEATURE FUSION

Types	2P2F-fuse			2P2F-assoc5			2P2F-assoc10		
	PR	RC	F1	PR	RC	F1	PR	RC	F1
Baidu image	0.97	0.98	0.97	0.97	0.97	0.97	0.93	0.97	0.95
Baidu video	0.98	0.98	0.98	0.84	0.97	0.90	0.99	0.99	0.99
Bilibili radio	0.90	0.85	0.87	0.89	0.88	0.88	0.79	0.67	0.73
Bilibili live	0.85	0.90	0.88	0.88	0.87	0.88	0.71	0.81	0.76
Toutiao image	1.00	0.98	0.99	0.97	0.96	0.96	0.98	0.93	0.96
Toutiao live	1.00	1.00	1.00	0.99	0.88	0.93	0.99	1.00	1.00
Netease cloud audio	0.96	0.98	0.97	0.99	0.89	0.94	0.95	0.99	0.97
Netease cloud video	0.98	0.97	0.98	0.95	0.72	0.82	0.99	0.97	0.98
Himalaya audio	1.00	0.98	0.99	1.00	0.99	1.00	0.99	1.00	0.99
Himalaya radio	0.98	1.00	0.99	1.00	1.00	1.00	0.99	1.00	1.00
Average	0.96	0.96	0.96	0.95	0.91	0.93	0.93	0.93	0.93

secure data transmission phase. Furthermore, the performance of fused features for classification is better than that of individual features.

The content-type classification results are presented in Table VI indicate that the length information of the TCP data segment in the secure data transmission phase is more suitable than the information of the protocol extension field in terms of content-type classification. This finding confirms the correctness of our feature selection method. Moreover, the two-phase feature fusion leads to an improvement in the accuracy of content type classification, which is similar to the improvement observed in application classification. The precision, recall, and F1-Measure of feature fusion are also higher than those of individual features.

2) *Effectiveness of Feature Fusion for Multi-Task*: 2P2F-assoc10 uses the two features mentioned above for application-content type recognition directly and selects the result with higher confidence of the two features.

The accuracy of the comprehensive classification obtained by combining different classification tasks using feature fusion is higher than that obtained by direct combination application type and content type and higher than that obtained by direct combination application-content type (Probability addition). This finding highlights the effectiveness of feature fusion for multi-task classification. Moreover, the superiority of application-content type classification based on feature fusion is confirmed by the evaluation of other indicators presented in Table VII.

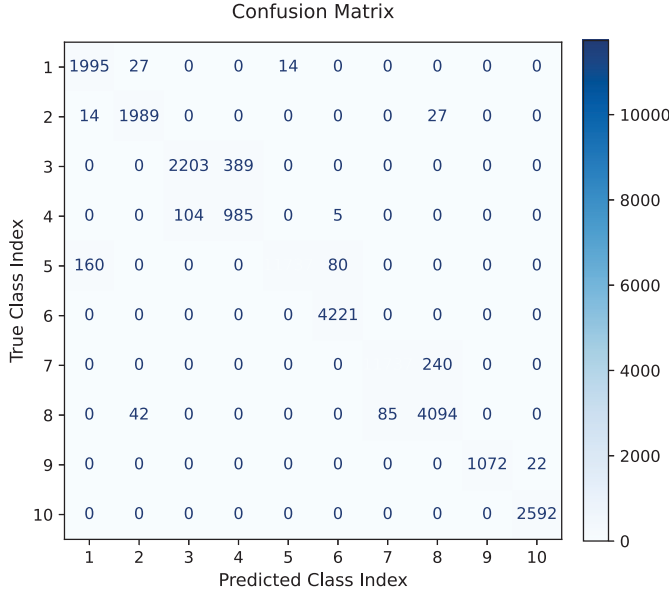


Fig. 8. Confusion matrix of the proposed 2P2F method. The class indices correspond to: 1) Baidu image, 2) Baidu video, 3) Bilibili radio, 4) Bilibili live video, 5) TouTiao image, 6) TouTiao live video, 7) Netease cloud audio, 8) Netease cloud video, 9) Himalaya audio, and 10) Himalaya radio.

To further analyze the detailed performance of the 2P2F-fuse model on application-content classification task, we plotted its confusion matrix on the 2P2F Website test set, as shown in Fig. 8. The diagonal elements of the matrix represent the number of correctly classified samples, while the off-diagonal elements show the specific confusion between different classes. It is clear from the figure that the model performs exceptionally well on most classes, such as “TouTiao live” and “Himalaya radio”, which achieved near-perfect classification. The main confusion occurs between semantically similar classes. For example, a small number of “Bilibili radio” samples were misclassified as “Bilibili live”, and there was also minor confusion between “Baidu image” and “Baidu video”. This aligns with the relatively lower F1-scores for the “Bilibili radio” and “Bilibili live” classes presented in Table VII.

F. Robustness Analysis

Recent systematization efforts (e.g., the SoK by Wickramasinghe et al. [42]) highlight that modern encrypted traffic may be subject to padding and traffic-shaping mechanisms, which can substantially weaken classifiers that rely on packet/record length statistics. Since the transmission branch of 2P2F models the TCP segment length sequence with a Transformer, it is necessary to assess how sensitive the overall framework is to length obfuscation.

1) *Experimental Setup*: We perform a controlled simulation on the 2P2F-Website dataset by perturbing the test-set TCP segment length sequences to emulate different degrees of length obfuscation, while keeping the handshake inputs unchanged. This isolates the effect of padding on the transmission-phase features and allows us to evaluate the benefit of two-phase fusion under adverse conditions. Specifically,

TABLE VIII
PERFORMANCE COMPARISON (F1-SCORE) UNDER
LENGTH-OBUSCATION SCENARIOS

Model Variant	Original (No Padding)	Random Padding	Block Padding (256 Bytes)
2P2F-transmission	0.90	0.89	0.59
2P2F-fuse (Ours)	0.96	0.96	0.88

for each TCP segment length L , we construct a padded length L' using one of the following strategies:

- **Random Padding**: $L' = L \cdot (1 + \delta)$, where $\delta \sim \mathcal{U}(0.10, 0.30)$. This emulates mild and variable padding/noise that partially perturbs length statistics.
- **Block Padding**: $L' = \lceil L/256 \rceil \cdot 256$. This emulates aggressive *block-quantization* shaping that coarsens length resolution and makes different flows more similar from a length-statistics perspective.

Table VIII indicates that the transmission-only variant is sensitive to block padding. When lengths are rounded to 256-byte blocks, fine-grained length differences across traffic classes are removed, making the resulting length sequences more similar and therefore harder to separate, which leads to a large F1 drop ($0.90 \rightarrow 0.59$).

In contrast, the fused model remains robust because it can still rely on handshake-phase TLS metadata (extension fields) that is not affected by transmission-phase length obfuscation. Consequently, even when length cues are degraded, the handshake branch provides complementary information, allowing 2P2F-fuse to retain strong performance ($F1 = 0.88$).

2) *Discussion on Contextual Overfitting*: The above padding stress-test complements recent SoK findings on encrypted traffic classifiers, which caution that performance may be driven by contextual artifacts (e.g., collection environment, implementation behaviors, or network conditions) rather than intrinsic application/content-type regularities. From this perspective, padding and traffic shaping can be viewed as one practical source of context shift that directly perturbs length-based side channels.

To reduce context dependence, we deliberately exclude highly variable features such as packet inter-arrival time and avoid using IP addresses directly due to the instability introduced by CDN and load balancing. Nevertheless, we acknowledge that side-channel-based methods may still be sensitive to broader context shifts (e.g., device/browser versions, network paths, and server deployments). We therefore treat cross-context evaluation (e.g., train/test splits across time or network environments) as an important direction for future work and include this limitation to contextualize applicability.

G. Time Overhead Analysis

Time overhead is a critical metric for practical, real-time traffic classification scenarios. To evaluate the feasibility of our 2P2F scheme, we conducted an experiment to measure its processing latency per network flow, and compared it with DE-GNN, a representative SOTA method that also requires complex feature engineering.

TABLE IX
TIME OVERHEAD COMPARISON (AVERAGE PER FLOW)

Time Component (ms)	2P2F (Ours)	DE-GNN
Preprocessing & Feature Extraction	4.82	25.41
Model Inference	0.91	2.15
Total Average Overhead	5.73	27.56

The analysis was performed on our testbed, which is equipped with an Intel Xeon Silver 4214 CPU and an NVIDIA V100 GPU. We measured the average time for two main stages:

- **Preprocessing & Feature Extraction:** For our 2P2F, this includes parsing the flow from raw packets, identifying the handshake and secure data transmission phases, generating the protocol extension field image (16×28), and extracting the TCP segment length sequence (first 32 segments). For DE-GNN, this involves the complex construction of the Traffic Interaction Graph (TIG) for each flow.
- **Model Inference Time:** This is the computation time required by the respective pre-trained models (2P2F-fuse vs. DE-GNN) to process the features and produce a classification result.

The results of this comparison are summarized in Table IX. Our 2P2F model demonstrates high efficiency. The combined preprocessing and feature extraction, which involves lightweight parsing and sequence extraction, takes only 4.82 ms on average. The model inference is also very fast, at 0.91 ms.

In contrast, the DE-GNN model incurs a significantly higher overhead. As hypothesized, its preprocessing stage, which requires constructing a complete Traffic Interaction Graph for each flow, is computationally intensive, averaging 25.41 ms. While its GNN inference time is also efficient (2.15 ms), its total overhead is nearly 5 times that of our method.

H. Summary

Based on the experimental findings, we can conclude that the grayscale image of the protocol extension fields during the handshake phase is suitable for application-type classification, while the TCP segment length sequence during the secure data transmission phase is most suitable for content-type classification. The features from both phases complement each other, which is why we have fused them together. By doing so, we have improved the accuracy of both application and content type classification. Furthermore, our feature fusion approach has achieved the best results for application-content type classification, thus confirming the effectiveness and superiority of the proposed 2P2F scheme.

VI. CONCLUSION

In this paper, we propose 2P2F for classifying encrypted network data flows by leveraging the distinct characteristics of different network communication phases and effectively fusing the extracted features. The 2P2F integrates a CNN and a

transformer model. The CNN analyzes the encryption protocol extension field during the handshake phase, extracting features relevant to the application type. Meanwhile, the transformer model processes the sequences of TCP segment lengths in the secure data transmission phase, extracting features related to the content type. Finally, an attention mechanism fuses the features extracted from both phases, facilitating the classification of the encrypted network data flow. The effectiveness of the 2P2F method is validated through comparative experiments and feature analysis, demonstrating its superior performance.

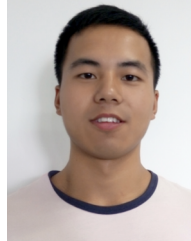
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